

Neutron generation dynamics inside a MA-class dense plasma focus Z-pinch

Cite as: Phys. Plasmas **32**, 033105 (2025); doi: [10.1063/5.0253547](https://doi.org/10.1063/5.0253547)

Submitted: 17 December 2024 · Accepted: 22 February 2025 ·

Published Online: 17 March 2025



View Online



Export Citation



CrossMark

C. Goyon,^{1,a)} A. J. Link,¹ O. B. Drury,¹ A. E. Youmans,¹ J. Angus,¹ S. Jiang,¹ S. Rocco,¹ S. F. Chapman,¹ C. M. Cooper,¹ M. McMahon,¹ A. Jibodu,^{1,2} D. Max,³ R. Anaya,¹ P. C. Campbell,¹ J. Park,¹ B. Shaw,¹ J. Sin,¹ and A. E. Schmidt¹

AFFILIATIONS

¹Lawrence Livermore National Laboratory, 7000 East Av., Livermore, California 94550, USA

²Stanford Plasma Physics Laboratory, Stanford University, Stanford, California 94305, USA

³Nevada National Security Site, North Las Vegas, Nevada 89030, USA

Note: This paper is part of the Special Collection: Papers from the 66th Annual Meeting of the APS Division of Plasma Physics.

^{a)}Author to whom correspondence should be addressed: goyon1@llnl.gov

ABSTRACT

Dense plasma foci (DPFs) are appealing as energy efficient sources of short pulses of ions, neutrons, and x rays. The output of these sources is expected to scale with input current (I^4), but has been shown to drop at the MA level [S. K. H. Auluck, “On the failure of neutron yield scaling in the dense plasma focus,” Phys. Plasmas **30**, 080701 (2023)]. New results on the MegaJoule Neutron Imaging Radiography DPF showed neutron yield production in agreement with the input current scaling beyond the previously observed drop. This work provides insight into the pinch formation on a DPF and reports on the two different mechanisms leading to neutron generation inside a DPF using a combination of kinetic simulations and experimental data. A combination of particle-in-cell (PIC) and 1D shock theory results are used to describe the pinch formation and disassembly and the corresponding thermonuclear and beam-target mechanisms. The temporal evolution of the pinch column predicted by the PIC simulations shows qualitative agreement with the experimental data from plasma photon emission as well as temporal neutron pulse shapes. In MJ-class DPFs, both thermonuclear and beam-target mechanisms can occur over the course of the implosion and contribute to the total neutron production. Hence the neutron source size of a DPF will change throughout the implosion. Experimental neutron radiographs show the increase in source size as the pinch breaks apart, in agreement with simulation’s prediction.

© 2025 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC) license (<https://creativecommons.org/licenses/by-nc/4.0/>). <https://doi.org/10.1063/5.0253547>

I. INTRODUCTION

Dense plasma focus (DPF) Z-pinchs have been used as neutron sources for ~60 years.¹ DPFs are a well-studied source of short (<100 ns) pulses of ions, neutrons, and x rays. DPF-based platforms are attractive because they exhibit high conversion efficiency of stored energy into radiation, and can typically be constructed at a lower cost than traditional accelerators for equivalent output. The scaling of yield vs input current and energy is also favorable, but has previously been shown to saturate for some devices at the MA and MJ level.² However, there have only been a few DPF with MA currents and even fewer driven by MJ pulsed power banks to study this limit. New results on the MegaJoule Neutron Imaging Radiography (MJOLNIR) DPF device demonstrate that design changes, led by experimentally validated simulations, increase the neutron yield by 30× since MJOLNIR’s initial discharge.³ MJOLNIR has demonstrated some of

the highest neutron yields achieved with a deuterium DPF at 8×10^{11} . The notable DPFs in this class are 0.42 MJ 2.3 MA LANL DPF-6 (1974),⁴ 0.5 MJ 4.6 MA Poseidon DPF (1987),⁵ 0.75 MJ 2 MA Verus research DPF,⁶ 1 MJ 2 MA PF-1000 DPF (1998),⁷ and the 2 MJ 6 MA Gemini DPF (2009).⁸ To our knowledge, the Poseidon, Verus, and PF-1000 have reached yields of 4.6×10^{11} , 4.6×10^{11} , and 2×10^{11} , respectively, while LANL DPF and Gemini have reported achieving $1\text{--}2 \times 10^{12}$ D-D yields in non-peer reviewed reports. These higher yields have been achieved after various iterations of the anode shape suggested by MHD-kinetic modeling techniques of the implosion.

This work presents a holistic comparison of experimental observations to a DPF modeling paradigm developed at Lawrence Livermore National Laboratory. A variety of models are used to understand DPF discharges, the most computationally intensive of which are pushing the boundaries of state-of-the-art physics fidelity. We start

by detailing the MJOLNIR DPF that was built to deliberately exceed the 3 MA threshold aiming to continue to produce an increasing number of neutrons with increasing current. The temporal evolution during the final part of the discharge is characterized by a simplified expansion model and synthetic diagnostics from particle-in-cell (PIC) simulations. The insights gained from the simulations are benchmarked against experimental results to show that from implosion to long after stagnation, multiple neutron generation events must be considered to explain the device output. Finally, the temporal evolution of neutron generation is tied to the time-evolving neutron source size showing that careful considerations of probing time may help push the boundaries of present limitations of DPFs as sources.^{9,10}

II. THE MJOLNIR PLATFORM

A. The MJOLNIR dense plasma focus

The MJOLNIR project aims to characterize the neutron yield, source size, and pulse shape generated by a DPF with 2 MJ of stored energy and provide a comprehensive study of the viability of DPFs for dynamic neutron radiography applications.¹¹ The MJOLNIR experimental platform relies on 6 dry air insulated, pressurized capacitor towers containing a total of 24 single-stage Marx generators and is able to deliver up to ~ 4.5 MA of current to the experimental head. Each tower holds eight capacitors and four rail gap switches repurposed from the ATLAS pulsed power facility.¹² A single Marx generator can produce voltages between 60 and 100 kV and stores up to 83 kJ, for a total maximum stored energy of 2 MJ. To this date, the machine has been commissioned up to 3.8 MA, which corresponds to a charge voltage of 80 kV.

The current is delivered from the capacitor bank to the experimental head via a transmission plate that connects the 168 cables coming from the towers to the central electrode inside the vacuum vessel. Figure 1 shows the vacuum vessel containing the electrodes and the diagnostics used to characterize the discharge. A detailed description of the electrodes can be found in Schmidt *et al.*³ They consist of a central 228.6 mm diameter copper anode and a set of 24 cathode rods surrounding it. The profile of the anode is made of three sections. An initial flat section is followed by a tapered part with a 15° angle. The anode ends in a flat section with a 2.5 cm radius, called the implosion radius, R_{imp} . The anode is hollowed at the tip by voiding the copper material along the longitudinal axis of the anode to minimize copper sputter.¹³ The anode and cathode are separated by a 12.5 mm thick quartz insulator covering ~ 4 cm of the base of the anode. The vacuum chamber is filled with deuterium gas with pressure P_{fill} , ranging from 5 to 16 Torr for the discharges presented in this work.

Since its first discharge, MJOLNIR has continuously expanded its diagnostic suite and a subset is described in this work to highlight the physics of the final stage of the discharge. A Rogowski coil located at the base of the central anode measures the current flowing to the experimental head. The light emission from the plasma sheath is captured by a fast framing camera sensitive to visible wavelength and taking 16 exposures of ~ 5 ns duration. The neutron yield is characterized using three different activation reactions from Be, Y, and Br. The beryllium activation is measured using two separate photomultiplier tubes¹⁴ and has been absolutely calibrated at the Sandia Ion Beam Laboratory. LaBr detectors with yttrium caps¹⁵ have been deployed at different polar angles around the source, after being cross-calibrated against the Be detector, to assess the anisotropy of the

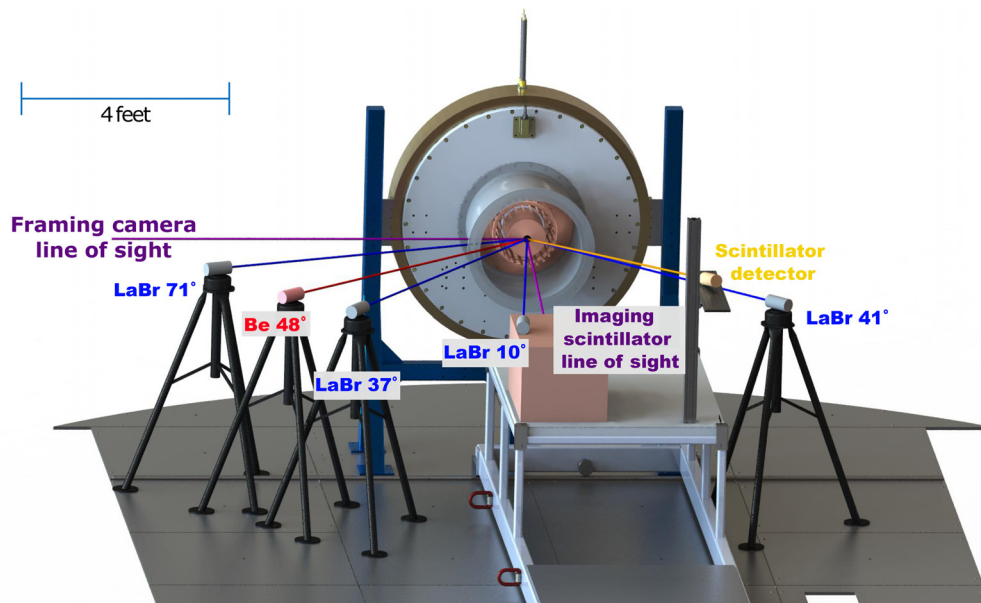


FIG. 1. Rendering of the inside of the MJOLNIR experimental chamber with the different diagnostic lines of sight. The inside of the chamber depicts the two coaxial electrodes: the central anode with a hollowed tip and the cathode rod cage around it. The insulator separating them is also shown in white. The intersection of the lines of sight close to the tip of the anode points to where the pinch occurs at the end of a discharge. Further down the axis, the enclosure containing the imaging scintillator for neutron radiography is shown. Each LaBr detector is labeled in blue along with its polar angle (considering the pinch as the origin). The Be activation detector is shown in red along with its polar angle. Note that the shape of the different activation and scintillator detectors does not correspond to their real shape.

source. Scintillator detectors for time-of-flight measurements have been placed at two different distances from the source, 2.2 and 6.6 m, to measure the pulse shape. A dedicated fiber array scintillator placed 1.9 m away on-axis is connected to a CMOS camera to obtain a spatially resolved image of the source.

All the detectors have been timed with respect to each other to within 1 ns. Cables were measured using optical time domain reflectometry, photomultiplier tube's time response was also characterized using a 80 ps laser. The head Rogowski was calibrated using a network analyzer on a test cell that mimics its deployment in the anode-cathode geometry. Current flowing to the central electrode is calculated by using an average inductance and a cumulative trapezoidal integration of the data. Other detectors, not presented here, include Rogowski coils to measure voltage between anode and cathode, residual gas analyzer, fiber-coupled photodiodes and x-ray step-wedge filter. The MJOLNIR platform pairs this extensive suite of diagnostics with state-of-the-art modeling capabilities.

B. Modeling a plasma focus discharge

Kinetic DPF modeling^{16–18} only became available in the last decade with the development of large supercomputers capable of running simulations at the length and time scales necessary to capture the physics of each stage of a discharge. Simulations need to encompass the entire device to capture the impact of the insulator shape on the liftoff, interactions of the plasma with the cathode rods, and asymmetries in the implosion due to the taper and shape of the anode tip. Because neutrons are primarily generated via mechanisms¹⁹ where kinetic physics is inherent to the problem, such a description is required at the later stage of the implosion. However, some stages such as run-down as well as parts of the run-in and liftoff can be well described by fluid theory. The primary simulation code used is the Chicago²⁰ code, which allows for both fluid and kinetic physics in a single simulation. These simulations are driven by a transmission line circuit model²¹ of the full MJOLNIR pulsed power system. These simulations have features as small as 50 μm and processes that evolve on 100 fs times cales while requiring the modeling of the 10–100 cm scale electrodes for microseconds. Simulations are initialized as a fully ionized fluid¹⁸ at 0.1 eV with a thin 1 mm layer along the surface of the insulator initialized to 1 eV. The simulation is run in single fluid MHD²² until the plasma sheath reaches the anode tip. The simulation transitions from the fluid description to a quasi-neutral kinetic ion with a fluid electron description for the implosion phase. The kinetic description helps to account for finite ion Larmor radius and mean free path effects which can change the characteristics of the implosion.²³ The initial particles per cell at the conversion time from fluid to kinetic is 250, which due to the convergence typically exceeds 4000 particles per cell at stagnation. Typically, simulations are limited to a duration of 100 ns after stagnation due to computational expense, using 1×10^6 CPU hours in MHD-kinetic mode.

MJOLNIR pairs experiments with MHD-kinetic modeling and reduced-order models based on more computationally intensive ones. In the following, synthetic diagnostics based on MHD-kinetic simulations are compared to experimental measurements of the DPF plasma discharge to validate the modeling, providing insight into the physics of the neutron generation of a DPF while the reduced-order model is detailed to describe the pinch disassembly time.

III. PLASMA EVOLUTION DURING THE IMPLOSION

A DPF discharge can be characterized by four distinct stages. First, the high voltage applied to the electrodes generates field-emitted electrons from the cathode that ionize the gas en route to the anode. This forms a conducting channel that completes the circuit and allows current to flow. This current interacts with the self-generated magnetic field through the " $\vec{j} \times \vec{B}$ " Lorentz force and accelerates a plasma sheath away from the insulator, referred to as the "break down" and "liftoff" stages. Second, the sheath continues to be accelerated under the $\vec{j} \times \vec{B}$ force and "runs down" the anode in a manner analogous to a plasma rail gun while the current keeps increasing. Third, when the plasma sheath reaches the tapered section of the anode, the $\vec{j} \times \vec{B}$ force gains a radial component directed inward; the "run-in" stage starts as the plasma sheath converges toward the cylindrical anode's axis. As the radius of the current column decreases, the inductance of the load increases and the current flowing through the head starts to drop. Finally, plasma stagnates on-axis, forming a Z-pinch: a plasma with an axial current confined by its own magnetic field.

Figure 2 shows the temporal evolution of the current flowing to the experimental head and each stage of the discharge is symbolized by the numbers 1–4. It is worth noting that the current available at stagnation and not the peak current dictates the plasma conditions. In the following, we focus on the neutron generation mechanisms occurring during the fourth stage of the discharge: the pinch.

During the pinch, discrete transitions occur in the plasma column's behavior starting with an inward imploding sheath and ending in a broken apart central column. Figure 3 presents the spatial location of the plasma sheath near the tip of the anode using light emission as a function of time. Experimental data in the top row are compared to synthetic optical images from MHD-kinetic simulations in the bottom row, which include the effects of the finite spatial resolution and temporal gating of the optical system used in the experiment. Simulation images do not account for distortion caused by propagation through the plasma near the cathode rods. Synthetic optical images show a qualitative agreement with experimental data based on several features,

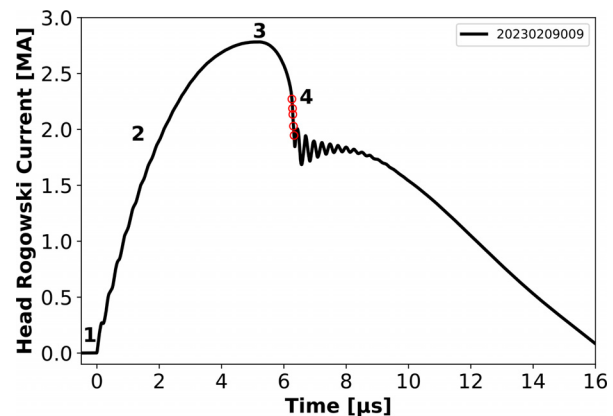


FIG. 2. Temporal evolution of the current flowing through the central anode ($R_{\text{imp}} = 2.5$ cm) using 0.75 MJ of stored energy and 7 Torr of D_2 gas-fill measured by the head Rogowski coil. Each stage of a DPF discharge is represented on the trace: (1) Insulator flashover, (2) run-down, (3) run-in, (4) pinch. The red circles symbolize the temporal location of each fast framing camera frame presented in Fig. 3.

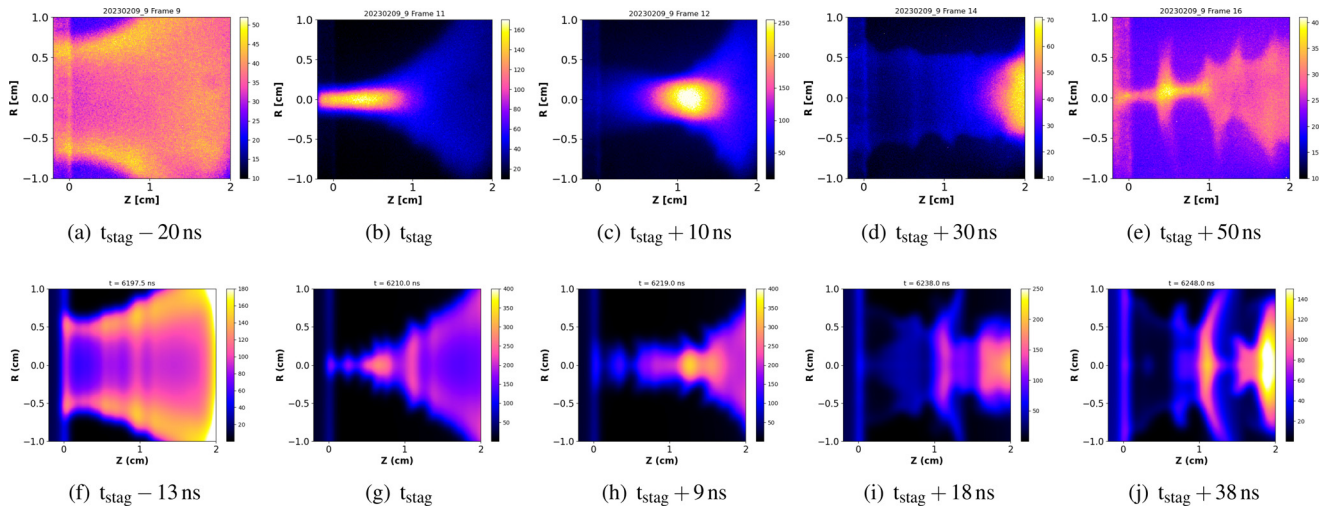


FIG. 3. Photon emission of the plasma sheath during the last stage of a DPF implosion with charge voltage of 60 kV and a gas-fill pressure of 9 Torr. Panels (a)–(e) show the experimental measurement of the fast framing camera with their timing reported with respect to stagnation. Panels (f)–(j) show the synthetic images generated from MHD-kinetic simulations. (a) $t_{\text{stag}} - 20$ ns, (b) t_{stag} , (c) $t_{\text{stag}} + 10$ ns, (d) $t_{\text{stag}} + 30$ ns, (e) $t_{\text{stag}} + 50$ ns, (f) $t_{\text{stag}} - 13$ ns, (g) t_{stag} , (h) $t_{\text{stag}} + 9$ ns, (i) $t_{\text{stag}} + 18$ ns, (j) $t_{\text{stag}} + 38$ ns.

more specifically the location of the plasma sheath's edges, the spatial scale of the plasma, the sequence of the different phases, and the location of the “hot” spots where the emission is brightest. The finer structures observed in the synthetic data can be attributed to the finite simulation resolution. The anode surface is approximated by a stair step profile seeding small plasma non-uniformities that are enhanced because of the azimuthal correlations in 2D simulations and are not expected to be fully representative of the experiment. At later times (~ 40 ns), kinetic instabilities are evolving slightly faster in the simulation, and there is some variation in the fine details of the pinch location which can be attributed to shot-to-shot fluctuations of the instability growth's location. Overall, all experimental discharges follow this same sequence of events with a similar time scale and show qualitative agreement with MHD-kinetic simulations conducted on a meter size box for several microseconds ($\approx 6 \mu\text{s}$).

The implosion velocity can be estimated from 1D shock theory²³ for a deuterium fill as

$$v_{\text{imp}} \approx 950 [\text{km/s}] \frac{I_{\text{imp}} [\text{MA}]}{R_{\text{imp}} [\text{cm}] \sqrt{P_{\text{fill}} [\text{Torr}]}}, \quad (1)$$

Using the parameters for the discharge presented in Fig. 3 and an implosion current $I_{\text{imp}} = 2$ MA, the implosion velocity, $v_{\text{imp}} = 259 \text{ km s}^{-1}$, is in good agreement with the experimental velocity inferred from the fast framing camera images²⁴ of $279 \pm 28 \text{ km s}^{-1}$. The incoming flow stagnates on-axis resulting in a hot compressed plasma. The average temperature at stagnation, $T_{\text{st}} = \langle T_e + T_i \rangle / 2$, can also be estimated from 1D shock theory for a deuterium plasma, with the fraction coupled to the electrons depending on collisionality of the plasma,²³

$$T_{\text{st}} \approx 21 [\text{keV}] \frac{I_{\text{imp}}^2 [\text{MA}]}{R_{\text{imp}}^2 [\text{cm}] P_{\text{fill}} [\text{Torr}]} \propto v_{\text{imp}}^2. \quad (2)$$

The stagnation column radius is correlated with the size of the implosion radius by the convergence ratio $CR = R_{\text{imp}} / r_{\text{st}}$. If the

temperature and density in the column are sufficient, thermonuclear neutrons may be generated. After stagnation, the plasma column expands outward at a velocity $v_{\text{exp}} \sim v_{\text{imp}} / 3$ until it decelerates and eventually stalls against the confining magnetic field at r_{exp} , which is approximately $2r_{\text{st}}$ in both experiments and simulations. For fixed stored energy in the capacitors and anode geometry, the expansion duration, τ_{exp} , increases with fill pressure,

$$\tau_{\text{exp}} [\text{ns}] = \frac{r_{\text{exp}} - r_{\text{st}}}{v_{\text{exp}}} \approx \frac{3R_{\text{imp}}}{v_{\text{imp}} \times CR} \approx 31.5 \frac{R_{\text{imp}}^2 [\text{cm}] \sqrt{P_{\text{fill}} [\text{Torr}]}}{CR \times I_{\text{imp}} [\text{MA}]}. \quad (3)$$

The deceleration causes a rippling of the outer surface due to the generation of hydrodynamic instabilities.²⁵ These ripples seed magneto-hydrodynamic instabilities that cause the plasma column to break apart in a manner analogous to a classic “ $m = 0$ ” instability.^{26–28} The time, τ_{m0} , for the instability to reach the axis can be estimated from the experimental parameters by assuming the instability grows at the sound speed from the position r_{exp} ,

$$\tau_{m0} [\text{ns}] = \frac{r_{\text{exp}}}{c_s} \approx 31.0 \frac{R_{\text{imp}}^2 [\text{cm}] \sqrt{P_{\text{fill}} [\text{Torr}]}}{CR \times I_{\text{imp}} [\text{MA}]}. \quad (4)$$

The temperature, T_{cs} , used to calculate the sound speed c_s at the onset of the instabilities assumes an adiabatic expansion and can be inferred from the temperature reached at stagnation $T_{\text{cs}} \sim (r_{\text{st}} / r_{\text{exp}})^2 T_{\text{st}}$. The approximate values in Eqs. (3) and (4) assume a pure deuterium working gas.

After the breakup, order of MV/mm self-generated electric fields are produced. They accelerate ions and electrons to supra-thermal velocities and in opposite directions. The electrons propagate primarily toward the anode where they generate x-ray radiation via bremsstrahlung. Ions are accelerated toward the plasma assembled downstream of the pinch region and generate neutrons via beam-target interaction,

which is the most common neutron production mechanism inside a DPF. This is especially true for lower stored energy devices where the implosion velocity is not sufficient to reach the high temperature required to produce thermal neutrons during the stagnation phase. Unlike most DPFs, the plasma temperature at stagnation can be on the order of several keV on MJOLNIR, allowing for thermonuclear fusion to generate a non-negligible part of the total neutron yield. Therefore, multiple neutron generation events will occur during a single discharge: thermonuclear at stagnation and beam-target events corresponding to pinch disruptions. Further investigation of the presence of multiple neutron generation events can be done by comparing expected neutron pulse shape from MHD-kinetic simulations and experimental measurements.

IV. NEUTRON PULSE SHAPE AND YIELD EVOLUTION

In this section, we compare the temporal evolution of the radiation emitted by the DPF, x rays and neutrons, from MHD-kinetic modeling to experimental data. Then, we correlate the pulse shape measured on the scintillator detectors with the timing of stagnation and pinch disruption images measured by the fast framing camera. All of the data presented in this work were obtained with the Marx generators charged to 60 kV, corresponding to 735 kJ of stored energy, delivering on average a ~ 2.8 MA peak current and ~ 2.1 MA at the time of stagnation.

A. Synthetic pulse shape

Kinetic simulations provide the neutron emission associated with stagnation and beam-target events, and their relative timing. Figure 4 presents the neutron pulse shape at the source location. Neutrons are emitted in successive peaks that can be attributed to different phases of the implosion. The first peak, labeled 1 in red, starts at $t = 0$ ns and corresponds to neutron emission at stagnation and is inferred to be primarily thermonuclear. A second emission burst occurs ~ 5 ns after stagnation, associated with the first pinch disruption observed in the simulations. This event is located inside the hollowed region of the anode, outside the field of view of the optical diagnostic. A second

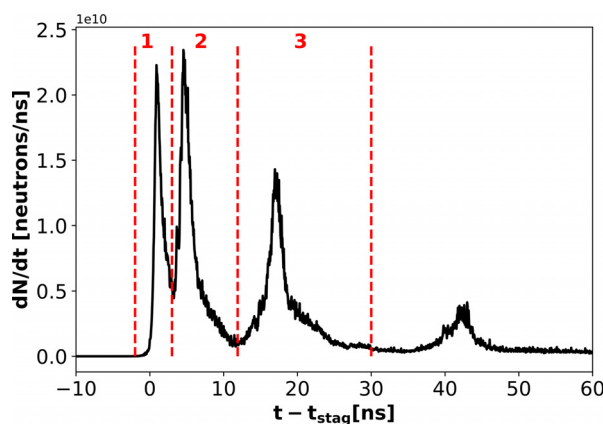


FIG. 4. Neutron birth time history from MHD-kinetic simulations of a 15° tapered anode for 60 kV charge voltage with 9 Torr of deuterium. Red temporal windows numbered 1, 2, and 3 each represent a neutron generation event. Window 1 corresponds to thermonuclear neutrons at stagnation and windows 2 and 3 correspond to beam-target events.

beam-target event, the third peak, is associated with a plasma disruption in front of the tip of the anode as presented in the right-most panel of Fig. 3. Finally, a third disruption leads to the last beam-target event producing neutrons. It is relatively weak compared to the other events as a significant amount of the energy in the device has been dissipated with the previous disruptions.

To compare the MHD-kinetic pulse shape to our scintillator detector measurements, we need to take into account the widening of the pulse with propagation distance and x-ray emission during the discharge. Each beam-target event is accompanied by an x-ray pulse coming from the electrons hitting the copper anode. Because electron collisions inside the copper are not modeled, we used the neutron pulse timing to generate peaks whose time-of-flight is consistent with photons emitted at each beam-target event. These mock x-ray peaks have been added to the synthetic time traces shown in Fig. 5 along with the experimental measurements for each detector location. Stagnation temperatures can reach several keV, which will produce x rays. However, the thickness of the aluminum chamber wall prevents photons with energies lower than 62 keV from reaching our detectors. Taking into account the 9 mm of additional lead shielding in front of the scintillator detectors and the fact that we are using low-Z deuterium gas, we expect the x-ray signature from the stagnation phase to be negligible and below the detection threshold compared to the bremsstrahlung emission from suprathermal electrons hitting the anode. For comparison, the experimental measurements are also plotted in the figure as solid lines for each respective distance. Experimental and synthetic data exhibit three peaks at ~ 10 , ~ 40 , and ~ 65 ns. These peaks are co-timed within 15 ns for all detector distances, which is consistent with time-of-flight at the speed of light. The observation of three x-ray peaks is also consistent with synthetic data and their timing agreement corroborates the occurrence of multiple beam-target events during a single MJOLNIR discharge as seen in the kinetic simulations.

A 2.45 MeV neutron should arrive at the first detector location 96 ns after being generated at the source, which corresponds to features observed after 100 ns. However, the experimental data show a single broad peak encompassing the two distinct ones seen in the simulated

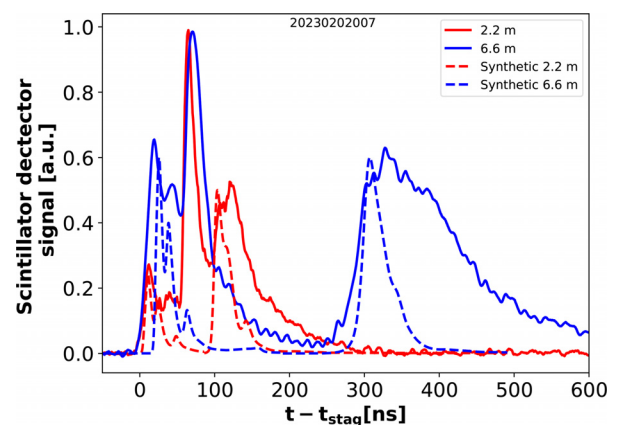


FIG. 5. Radiation pulse shape from a MJOLNIR discharge. Solid lines represent experimental data and dashed lines represent synthetic data generated using MHD-kinetic simulations for a 6 Torr, 60 kV discharges. Red lines correspond to measurements at 2.2 m and blue lines to 6.6 m.

traces. Further propagation to 6.6 m shows a similar trend: times-of-flight are consistent with rise times but experimental data exhibit a slow decaying peak compared to the simulations. The energy spread associated with each beam-target event is on the order of several hundreds of keV with maximum energy of ~ 4 MeV. This effect and the time smoothing due to the detector's temporal response are included in the synthetic traces but do not lead to sufficient smearing to match the experimental data. A possible explanation for the difference is the signal coming from neutrons scattering off of the experimental room and equipment inside, such as the vacuum chamber and the electrodes, which is not accounted for in the synthetic traces. Finally, it should be noted that a DPF implosion relies on instability growth which is intrinsically varying shot-to-shot. We use this single example to identify features that can be tracked over several discharges and provide overall trends.

B. Stagnation and breakup timing

Experimentally, we have accurately identified and tracked the first two x-ray peaks on the scintillator detector. Figure 6 compares the time difference, Δt , between the first two x-ray peaks to the time difference between the stagnation and pinch disruption frames (shown in Fig. 3). The time differences measured with each detector are in qualitative agreement as they both increase with the gas-fill pressure. The x-ray differences are consistently smaller, by an average of 16 ns. Such a difference can be attributed to the process generating the first x-ray pulse being delayed compared to the stagnation frame. The first x-ray peak is expected to be correlated with the first beam-target event occurring inside the hollowed tip while the framing camera tracks stagnation. The synthetic pulse shape predicts the first beam-target event will occur ~ 10 ns after stagnation, which is consistent with the reported measurements.

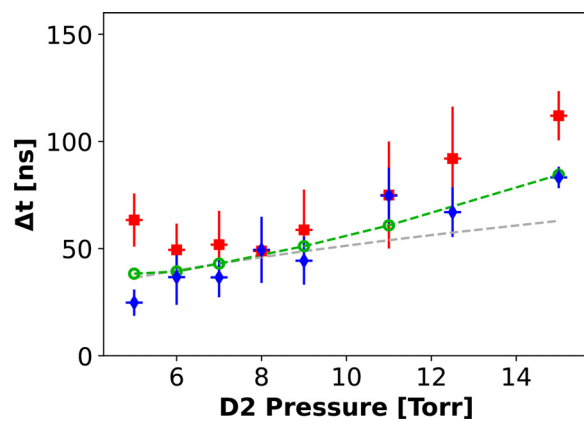


FIG. 6. The red squares correspond to the time difference between the frames showing the stagnation and the first pinch disruption outside the hollowed tip of the anode observed on the framing camera. The blue diamonds represent the temporal separation between the first two x-ray peaks experimentally measured using scintillator detectors. Each point corresponds to an average over at least four distinct discharges and the error bars represent the standard deviation. The dashed green line represents the sum of τ_{exp} and τ_{m0} as a function of the gas-fill pressure using the experimentally measured implosion velocity (solid points). The gray dashed line shows the same scaling assuming a constant implosion current.

Both Δt can be compared to the model for the time evolution of the implosion described in Sec. III. The dashed gray line shows the implosion characteristic time scaling as the square root of the density when using a fixed implosion current and a fixed $CR = 10$ inferred from the experimental data [Fig. 3(b)]. There is reasonable agreement with the experimental data at low pressures but the model predicts a faster disassembly above 10 Torr. This deviation is attributed to the increase in parasitic current losses at higher pressure, which has been published previously.²⁴ Parasitic currents divert some of the current measured entering the device from reaching the tip of the anode where the implosion occurs, leading to lower implosion currents and longer disassembly times. While the implosion current is not resolved experimentally, the parasitic current effect can be accounted for using the measured implosion velocity. Each point on the green dashed line shows the adjusted Δt values using the measured implosion velocities. Some pressures did not have high enough temporal resolution data on the framing camera to allow implosion velocity measurements. The adjusted model reproduces the experimental Δt at all pressures. The presence of multiple neutron generating events within a single MJOLNIR discharge is consistent with the measured pulse shape, the observation of multiple x-ray peaks as well as their temporal separation and can be associated with the stagnation and disruption of the pinch column. Further evidence of the presence of beam-target events can be found in the anisotropy of the neutron emission.

C. Time-integrated neutron anisotropy

An increased directionality of the neutron source could be direct evidence of changes in the ion beam properties. The kinetic energy of the accelerated ions is comparable to the D-D fusion Q (3.27 MeV for the $D + D \rightarrow n + {}^3\text{He}$). Therefore, a shift of the ion beam distribution toward higher energy would cause the neutrons to shift to higher energies and would increase the neutron fluence in the direction parallel to the beam. The temporal evolution of the neutron spectrum is shown in Fig. 7 using the data from MHD-kinetic simulations.

Each neutron generating event described in Fig. 4 is spectrally resolved. At stagnation, the spectrum is narrow around the 2.45 MeV energy expected in thermonuclear processes. As time evolves and the neutron generation shifts from thermal processes to beam-target, the

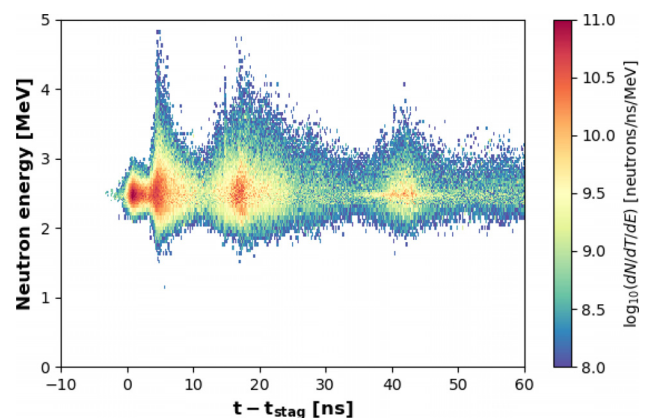


FIG. 7. Temporal evolution of the neutron energy spectrum from MHD-kinetic simulations of a 15° tapered anode for 60 kV charge voltage with 9 Torr of deuterium.

energy distribution broadens, showing a larger amount of higher energy neutrons, up to 5 MeV. The larger spectrum is observed for all three events associated with beam-target Sec. IV B.

The simulations show that thermal processes will not lead to significant energy spread, whereas the presence of beam-target events will be marked by broader spectrum. Therefore, the neutron emission from thermal processes will be more isotropic than the beam-target mechanism.

Experimentally, it is possible to estimate the anisotropy of the MJOLNIR DPF source by comparing activation detectors placed at two different angles: one close to the machine axis and one nearly perpendicular to it. The ratio of these two activation detectors placed at 10° and 70° from the longitudinal axis of the device is shown in Fig. 8. The on-axis/off-axis measurement is plotted as a function of the 4π -yield estimated using the Be detector at 45° and distinguishes between yttrium and bromine activation reactions. At low 4π -yield, all measurements are within the 10% error bar unlike high reference yield, where the on-axis measurements rises up to 80%–100% higher than the off-axis one. The bromine activation measurements report the same trend but the slope of the ratio increase is less pronounced with activation reaching up to 60%–70% higher values on-axis. Experimental measurements on MJOLNIR and other DPFs^{29–31} suggest neutron emission preferentially in the forward direction as the yield increases.

We conclude that over the pressure scan, higher yields are accompanied by higher anisotropy with higher yield reported in the forward direction suggesting an increasing beam-target contribution at higher yields. This observation is consistent with previous publication²⁴ showing that high performing discharges coincide with higher current coupling to the plasma in the final stages of the implosion and limited parasitic currents and to zeroth order, such effects would affect the beam-target mechanism more significantly than the thermal process.

V. NEUTRON SPOT SIZE TEMPORAL EVOLUTION

Neutron source sizes are commonly measured with a pinhole (two-dimensional measurement) or a slit (one-dimensional measurement) in cases where the source is extended in one direction or yield is low. We have selected a slit as the baseline measurement to maximize

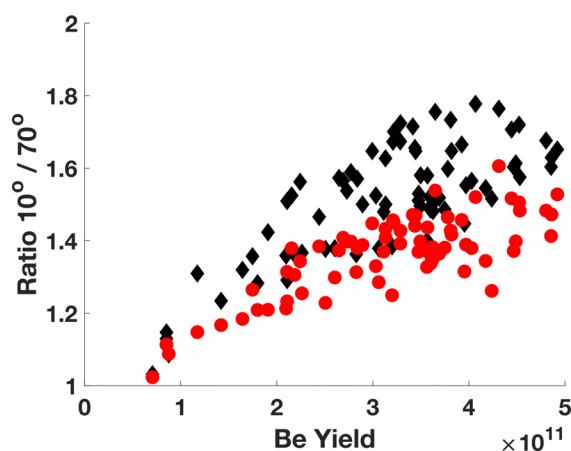


FIG. 8. Anisotropy factor estimated as ratio of neutron yields measured at 10° and 70° polar angle using the activation of yttrium (black diamonds) and bromine (red circles).

diagnostic sensitivity. The slit configuration has been fielded in multiple orientations for evaluation of any asymmetries in the “up–down” vs “left–right” distributions.

The neutron source imaging slit is comprised of two cylindrically faced surfaces (rolled edges) facing each other with a small separation. Due to the penetrating nature of neutrons, the choice of separation and surface radius is a compromise. The source resolution is $r_s = \sqrt{r_d^2 + w_{eff}^2} \times m^{-1}$, where r_d is the detector resolution, $m = L_s/L_d$ is the magnification and w_{eff} is the effective slit width (50% attenuation level), which takes into account the reduced attenuation of the edge due to curvature. A slit comprised of two rolled edges with “small” radii in contact will have a narrow region where neutrons penetrate and are recorded at the detector. We have selected a surface radius of 500 mm, which gives us the ability to measure spatial resolutions over a wide range of yields while maintaining sufficient field-of-view at the source and imaging scintillator planes to cover potential variations in neutron source position and size. We utilize a face separation of 1.0 mm and the thickness of the rolled edges was chosen to be ~ 4.5 times the mean free path for a 2.5 MeV neutron to provide good contrast.

The neutron imager is based on a design deployed on the National Ignition Facility.^{32–34} Neutrons interact with a $7.5 \times 7.5 \text{ cm}^2$, 5.08 cm-thick fiber bundle array scintillator. Each fiber is $250 \mu\text{m}$ diameter and made of plastic (BCF-99-55). Typical settings for imaging with the second iteration of the imager use a 50 ns time-gate for the first MCP and a 10 ms exposure for the camera.

We show in Fig. 9 four radiographs of a slit taken 2 m away from the source and at different times during the neutron pulse shown as the solid red trace in Fig. 5. The left-most image corresponds to the earliest time within the neutron pulse; the exposure starts at the beginning of the neutron pulse and stops when peak signal is reached. We observe a centered vertical band where the signal is the brightest and corresponding to neutrons going through the slit. The rest of the image is much darker and constitutes the shadow of the two rolled edges. From left to right, the delay of the imager time-gate increases. The second image from the left is centered on the neutron peak, the third image starts at the peak, and the last image is recorded 20 ns after the neutron peak was reached. The brightness of the signal is highest in the second image, whose exposure includes most of the neutron peak. The signal brightness follows the trend we would expect: the greater the integrated neutron signal included in the exposure duration, the more signal gets through the slit. However, there is also a noticeable change in the slit sharpness, which is consistent with an increased source size. For each radiograph, we reconstruct the source size by assuming the radial profile of the source is Gaussian with a standard deviation σ and peak location μ . The fitting function is described by

$$\phi(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right]. \quad (5)$$

We conducted two timing scans at 7 and 11 Torr to quantify the time evolution of the spot size during the neutron pulse and assess the influence of gas-fill pressure. Figure 10 presents how the spot size increases as we delay the imager integration time relative to stagnation. As was the case for the data shown in Fig. 9, the earliest delay for which we have been able to obtain a viable radiograph (~ 80 ns) corresponds to the earliest timing, i.e., when the time-gate of the neutron

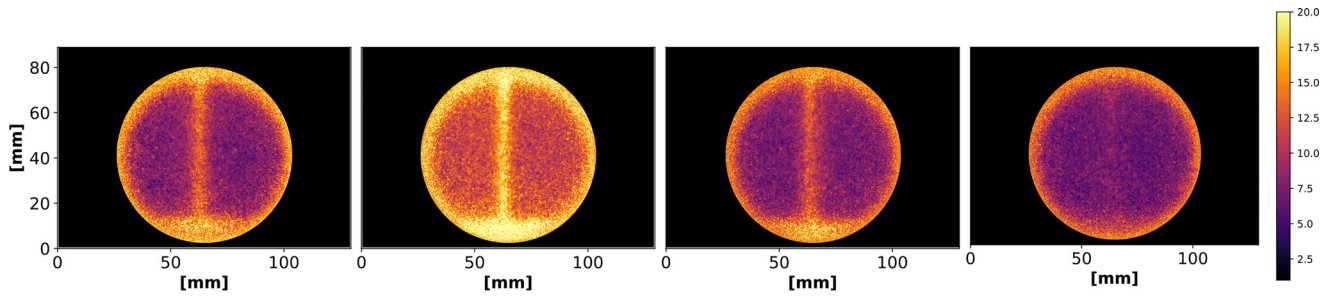


FIG. 9. Time evolution of a rolled edge radiograph throughout the neutron pulse for 6 Torr deuterium gas-fill and 60 kV charge voltage. Each image corresponds to a 50 ns-long exposure with start times of 94, 113, 120, and 140 ns after stagnation, increasing from left to right.

imager overlaps with the peak neutron signal. For both pressures, the inferred spot size increases as the time-gate is moved toward the end of the neutron pulse. The spot size increases from ~ 6 to >10 mm during the trailing edge of the neutron pulse. The magnitude of the spot size increase is $\sim 40\%$, which is much larger than the 5% blurring, estimated with MCNP,³⁵ coming from the scattered neutron signal. The same diagnostic was also used to characterize the MJOLNIR x-ray source size. We inferred that the x-ray source from this DPF configuration is much larger ($>10\times$) than the neutron size and that the x-ray and neutron pulses do not overlap in time at the imager distance.

MHD-kinetic simulations also provide the temporal evolution of the source size as seen in Fig. 11. From the successive peaks shown in Fig. 4, stagnation produces the smallest source size, on the order of the plasma column's diameter, then every successive neutron generating event contributes to increasing the source size over time as neutrons are generated in a larger volume. In the simulations, the radial profile of the plasma target that produces the neutrons has extended wings, which means the difference between the radius enclosing 80% of the neutrons and the radius enclosing 50% of the neutrons is up to seven-fold. This sensitivity to the target size combined with the uncertainty on the exact plasma conditions could explain why the simulations reproduce the overall trend but not the absolute numbers for the experimental neutron spot size. A small discrepancy between the experiment and the simulations in either the target conditions (temperature and density) or the ion beam energy distribution will be

compounded in the reactivity calculations and could cause the difference between experimental and simulated spot size.

Using the 80% contour seems to provide the best agreement with the experimental results as integration over the first three pulses leads to a 7 mm source diameter that increases to 9.5 mm when integrating over the whole neutron pulse duration. Without quoting a direct number for the expected spot size, MHD-kinetic simulations can be used to compare how the implosion radius would affect the expected source size, which is key for any radiography application.

VI. CONCLUSION

This work presents a MHD-kinetic description of the processes leading to neutron generation inside a dense plasma focus. The temporal dynamics of a discharge from the MJOLNIR dense plasma focus is detailed using 2D particle-in-cell simulations benchmarked against multiple experimental measurements of the discharge's visible, x-ray, and neutron emissions. Characteristic times of the discharge, such as implosion, stagnation-to-expansion, and plasma breakup times, are tracked using fast framing camera images of the current sheath and scintillator detectors for x-ray emission. Experimentally measured

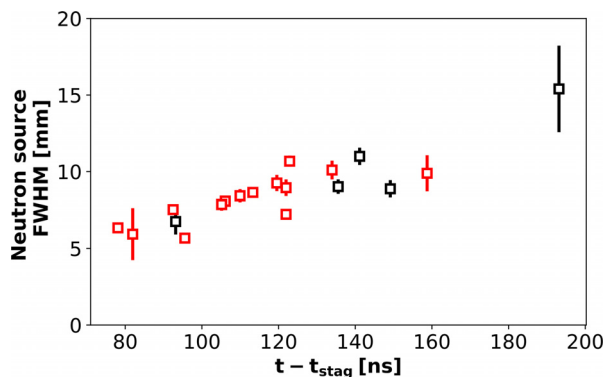


FIG. 10. Temporal evolution of the neutron source size with respect to stagnation time at 7 Torr (red) and 11 Torr (black) for a 60 kV charge voltage.

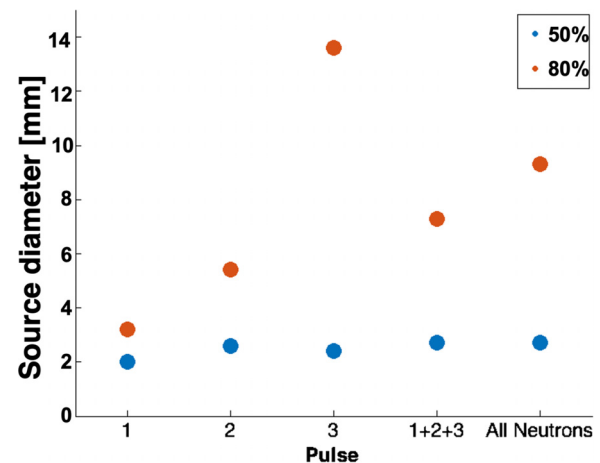


FIG. 11. Simulated neutron source size using MHD-kinetic simulations for different time integration windows including 1, 2, 3 pulses or the entire neutron pulse duration. Neutrons are characterized by their properties at generation or birth without any propagation effects. Blue dots represent the diameter including 80% of the neutrons generated and red dots the diameter where 50% of the neutrons are generated.

characteristic times are consistent with both PIC simulations and 1D shock theory's expectations over a wide range of gas-fill pressures. This corroborates the description presented in this paper where both thermonuclear and beam-target processes are contributing to neutron generation in the MJOLNIR DPF. Finally, the temporal evolution of the neutron source size measured with a time gated imager shows the neutron generating volume is increasing in size through the neutron emission, consistent with a plasma target that is disassembling during the neutron generation process.

ACKNOWLEDGMENTS

This work was performed under the auspices of the U.S. Department of Energy under Contract No. DE-AC52-07NA27344 with support from the DOE Office of Science Early Career Research Program under No. SCW1741.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

C. Goyon: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Writing – original draft (equal); Writing – review & editing (equal). **A. J. Link:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (supporting); Investigation (equal); Writing – original draft (equal); Writing – review & editing (equal). **O. B. Drury:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Writing – original draft (supporting); Writing – review & editing (supporting). **A. E. Youmans:** Formal analysis (equal); Writing – original draft (supporting); Writing – review & editing (supporting). **J. Angus:** Data curation (supporting); Investigation (supporting); Writing – review & editing (supporting). **S. Jiang:** Formal analysis (supporting); Writing – review & editing (supporting). **S. Rocco:** Formal analysis (supporting); Writing – review & editing (supporting). **S. F. Chapman:** Writing – review & editing (supporting). **C. M. Cooper:** Formal analysis (supporting). **M. McMahon:** Formal analysis (supporting). **A. Jibodu:** Formal analysis (supporting). **D. Max:** Investigation (supporting). **R. Anaya:** Investigation (supporting). **P. C. Campbell:** Formal analysis (supporting). **J. Park:** Formal analysis (supporting). **B. Shaw:** Formal analysis (supporting). **J. Sin:** Visualization (supporting). **A. E. Schmidt:** Funding acquisition (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

- ¹J. W. Mather and P. J. Bottoms, "Characteristics of the dense plasma focus discharge," *Phys. Fluids* **11**, 611–618 (1968).
- ²S. K. H. Auluck, "On the failure of neutron yield scaling in the dense plasma focus," *Phys. Plasmas* **30**, 080701 (2023).
- ³A. Schmidt, E. Anaya, M. Anderson, J. Angus, P. C. Campbell, S. F. Chapman, C. M. Cooper, O. B. Drury, L. Frausto, C. Goyon, S. Jiang, A. Jibodu, E. Koh, A. Link, D. Max, J. Park, S. R. Rocco, J. K. Walters, and A. E. Youmans, "MegaJoule neutron imaging radiography (MJOLNIR) dense plasma focus rebuild and high current experiments," *IEEE Trans. Plasma Sci.* **52**, 4906–4915 (2024).
- ⁴O. Zucker, W. Bostick, R. Gullickson, J. Long, J. Luce, and I. I. Sahi, "The design of a repetitively pulsed MegaJoule dense-plasma focus," Report No. UCRL-51872, 1975.
- ⁵H. Herold, A. Jerzykiewicz, M. Sadowski, and H. Schmidt, "Comparative analysis of large plasma focus experiments performed at IPF, Stuttgart, and IPJ, Świerk," *Nucl. Fusion* **29**, 1255 (1989).
- ⁶R. H. Dwyer, C. Willis, and M. Butcher, "Design of a 16-channel fiber-optic coupled radiation detection array for radiation asymmetry studies in a 750-kJ dense plasma focus," *Rev. Sci. Instrum.* **92**, 043557 (2021).
- ⁷H. Schmidt, A. Kasperczuk, M. Paduch, T. Pisarczyk, M. Scholz, K. Tomaszewski, and A. Szydlowski, "Review of recent experiments with the MegaJoule PF-1000 plasma focus device," *Phys. Scr.* **66**, 168 (2002).
- ⁸S. Auluck, P. Kubas, M. Paduch, M. J. Sadowski, V. I. Krauz, S. Lee, L. Soto, M. Scholz, R. Miklaszewski, H. Schmidt, A. Blagoev, M. Samuelli, Y. S. Seng, S. V. Springham, A. Talebitaher, C. Pavez, M. Akel, S. L. Yap, R. Verma, K. Kolacek, P. L. C. Keat, R. S. Rawat, A. Abdou, G. Zhang, and T. Laas, "Update on the scientific status of the plasma focus," *Plasma* **4**, 450–669 (2021).
- ⁹L. Tafoya, V. Geppert-Kleinrath, J. Allison, S. Baker, J. Bundgaard, M. Freeman, A. Hayes-Sterbenz, G. Jungman, H. Li, S. Li, D. Lowe, J. Tybo, and C. Wilde, "Spot size measurement of a deuterium-tritium dense plasma focus using neutron radiography," *Rev. Sci. Instrum.* **95**, 053501 (2024).
- ¹⁰N. Bennett, M. Blasco, K. Breeding, D. Constantino, A. De Young, V. DiPuccio, J. Friedman, B. Gall, S. Gardner, J. Gatling, E. C. Hagen, A. Luttman, B. T. Meehan, M. Misch, S. Molnar, G. Morgan, R. O'Brien, L. Robbins, R. Rundberg, N. Sipe, D. R. Welch, and V. Yuan, "Development of the dense plasma focus for short-pulse applications," *Phys. Plasmas* **24**, 012702 (2017).
- ¹¹A. Schmidt, E. Anaya, M. Anderson, J. Angus, S. Chapman, C. Cooper, O. Drury, C. Goyon, S. Hawkins, D. P. Higginson, I. Holod, E. Koh, A. Link, D. Max, M. McMahon, J. Mitrani, Y. Podpaly, A. Povilus, and D. Van Lue, "First experiments and radiographs on the MegaJoule neutron imaging radiography (MJOLNIR) dense plasma focus," *IEEE Trans. Plasma Sci.* **49**, 3299–3306 (2021).
- ¹²W. M. Parsons, C. C. Baldwin, E. O. Ballard, R. R. Bartsch, J. F. Benage, G. A. Bennett, D. W. Bowman, J. C. Cochrane, J. Creager, H. A. Davis, C. A. Ekdahl, R. F. Gribble, J. R. Griego, W. B. Hinckley, K. W. Hosack, R. J. Kasik, R. Keinigs, H. J. Kimerly, E. A. Lopez, A. Marquez, M. D. Monroe, S. A. Ney, D. D. Pierce, D. Platts, W. A. Reass, H. R. Salazar, G. M. Sandoval, D. W. Scudder, J. S. Schlachter, M. C. Thompson, R. J. Trainor, G. A. Valdez, R. G. Watt, G. A. Wurden, F. J. Wysocki, and S. M. Younger, "An overview of the atlas pulsed-power systems," in *11th IEEE International Pulsed Power Conference (Cat. No. 97CH36127)* (IEEE, 1997), Vol. 1, pp. 673–678.
- ¹³B. H. Shaw, S. Chapman, C. M. Cooper, C. Goyon, J. Angus, A. Link, D. P. Higginson, J. X. Liu, J. M. Mitrani, Y. A. Podpaly, A. Povilus, and A. Schmidt, "Maximizing neutron yields by scaling hollow diameter of a dense plasma focus anode," *J. Appl. Phys.* **124**, 233301 (2018).
- ¹⁴C. L. Ruiz, J. D. Styron, D. L. Fehl, K. D. Hahn, B. McWatters, M. A. Mangan, G. W. Cooper, J. D. Vaughan, G. A. Chandler, B. M. Jones, J. A. Torres, S. A. Slutz, D. J. Ampleford, M. R. Gomez, E. Harding, A. J. Harvey-Thompson, and P. F. Knapp, "Novel beryllium-scintillator, neutron-fluence detector for magnetized liner inertial fusion experiments," *Phys. Rev. Accel. Beams* **22**, 042901 (2019).
- ¹⁵A. E. Youmans, J. M. Mitrani, M. McMahon, A. T. Jibodu, C. S. Goyon, C. M. Cooper, B. L. Goldblum, T. A. Laplace, J. A. Brown, F. Conti, A. Williams, F. N. Beg, E. Ruskov, B. J. Levitt, A. R. Johansen, U. Shumlak, P. E. Tsai, and D. P. Higginson, "PANDA-FES: Portable and adaptable neutron diagnostics for advancing fusion energy science," *IEEE Trans. Plasma Sci.* **52**, 4833–4841 (2024).
- ¹⁶A. Schmidt, V. Tang, and D. Welch, "Fully kinetic simulations of dense plasma focus Z-pinch devices," *Phys. Rev. Lett.* **109**, 205003 (2012).
- ¹⁷A. Schmidt, A. Link, D. Welch, J. Ellsworth, S. Falabella, and V. Tang, "Comparisons of dense-plasma-focus kinetic simulations with experimental measurements," *Phys. Rev. E* **89**, 061101 (2014).
- ¹⁸A. Link, C. Halvorson, E. C. Hagen, D. V. Rose, D. R. Welch, and A. Schmidt, "Particle-in-cell modeling for MJ scale dense plasma focus with varied anode shape," *AIP Conf. Proc.* **1639**, 23–26 (2014).

- ¹⁹W. Stygar, G. Gerdin, F. Venneri, and J. Mandrekas, "Particle beams generated by a 6–12.5 kJ dense plasma focus," *Nucl. Fusion* **22**, 1161 (1982).
- ²⁰C. Thoma, D. R. Welch, R. E. Clark, D. V. Rose, and I. E. Golovkin, "Hybrid-PIC modeling of laser-plasma interactions and hot electron generation in gold hohlraum walls," *Phys. Plasmas* **24**, 062707 (2017).
- ²¹D. Hinshelwood, "BERTHA-A versatile transmission line and circuit code," NRL Memorandum Report No. 5185, 1983.
- ²²C. Thoma, D. R. Welch, and D. V. Rose, "Implicit highly-coupled single-ion Hall-MHD formulation for hybrid particle-in-cell codes," *Comput. Phys. Commun.* **261**, 107823 (2021).
- ²³J. R. Angus, A. J. Link, and A. E. Schmidt, "1D kinetic study of pinch formation in a dense plasma focus: Transition from collisional to collisionless regimes," *Phys. Plasmas* **28**, 010701 (2021).
- ²⁴A. Schmidt, C. Goyon, A. Link, C. Cooper, E. Anaya, M. Anderson, G. Bartolo, P. C. Campbell, S. Chapman, D. Max, M. McMahon, Y. A. Podpaly, and A. Povilus, "Measuring characteristic differences between high- and low-performing discharges on the MegaJoule neutron imaging radiography (MJOLNIR) DPF," *Phys. Plasmas* **29**, 062708 (2022).
- ²⁵M. R. Weis, P. Zhang, Y. Y. Lau, P. F. Schmit, K. J. Peterson, M. Hess, and R. M. Gilgenbach, "Coupling of sausage, kink, and magneto-Rayleigh-Taylor instabilities in a cylindrical liner," *Phys. Plasmas* **22**, 032706 (2015).
- ²⁶L. Gomberoff and M. Hernández, "Stationary convection in a cylindrical plasma," *Phys. Rev. A* **27**, 1244–1246 (1983).
- ²⁷J. P. Chittenden, R. Aliaga Rossel, S. V. Lebedev, I. H. Mitchell, M. Tatarakis, A. R. Bell, and M. G. Haines, "Two-dimensional magneto-hydrodynamic modeling of carbon fiber Z-pinch experiments," *Phys. Plasmas* **4**, 4309–4317 (1997).
- ²⁸M. G. Haines, M. S. Wei, F. N. Beg, and R. B. Stephens, "Hot-electron temperature and laser-light absorption in fast ignition," *Phys. Rev. Lett.* **102**, 045008 (2009).
- ²⁹F. Castillo, M. Milanese, R. Moroso, and J. Pouzo, "Evidence of thermal and non-thermal mechanisms coexisting in dense plasma focus D-D nuclear reactions," *J. Phys. D: Appl. Phys.* **33**, 141 (2000).
- ³⁰F. Castillo Mejía, M. Milanese, R. Moroso, and J. Pouzo, "Some experimental research on anisotropic effects in the neutron emission of dense plasma-focus devices," *J. Phys. D: Appl. Phys.* **30**, 1499 (1997).
- ³¹R. Aliaga-Rossel and P. Choi, "Experimental observations of the spatial anisotropy of the neutron emission in a medium energy plasma focus," *IEEE Trans. Plasma Sci.* **26**, 1138–1145 (1998).
- ³²M. D. Wilke, S. H. Batha, P. A. Bradley, R. D. Day, D. D. Clark, V. E. Fatherley, J. P. Finch, R. A. Gallegos, F. P. Garcia, G. P. Grim, S. A. Jaramillo, A. J. Montoya, M. J. Moran, G. L. Morgan, J. A. Oertel, T. A. Ortiz, J. R. Payton, P. Pazuchanics, D. W. Schmidt, A. C. Valdez, C. H. Wilde, and D. C. Wilson, "The national ignition facility neutron imaging system," *Rev. Sci. Instrum.* **79**, 10E529 (2008).
- ³³D. N. Fittinghoff, D. E. Bower, J. R. Hollaway, B. A. Jacoby, P. B. Weiss, R. A. Buckles, T. J. Sammons, L. A. McPherson, Jr., C. L. Ruiz, G. A. Chandler, J. A. Torres, R. J. Leeper, G. W. Cooper, and A. J. Nelson, "One-dimensional neutron imager for the Sandia Z facility," *Rev. Sci. Instrum.* **79**, 10E530 (2008).
- ³⁴F. E. Merrill, D. Bower, R. Buckles, D. D. Clark, C. R. Danly, O. B. Drury, J. M. Dzenitis, V. E. Fatherley, D. N. Fittinghoff, R. Gallegos, G. P. Grim, N. Guler, E. N. Loomis, S. Lutz, R. M. Malone, D. D. Martinson, D. Mares, D. J. Morley, G. L. Morgan, J. A. Oertel, I. L. Tregillis, P. L. Volegov, P. B. Weiss, C. H. Wilde, and D. C. Wilson, "The neutron imaging diagnostic at NIF (invited)," *Rev. Sci. Instrum.* **83**, 10D317 (2012).
- ³⁵J. A. Kulesza, T. R. Adams, J. C. Armstrong, S. R. Bolding, F. B. Brown, J. S. Bull, T. P. Burke, A. R. Clark, R. Forster, I. Arthur, J. F. Giron, T. S. Grieve, C. J. Josey, R. L. Martz, G. W. McKinney, E. J. Pearson, M. E. Rising, J. Solomon, S. Swaminarayan, T. Trahan, S. C. Wilson, and A. J. Zukaitis, "MCNP code version 6.3.0 theory and user manual," Report No. LA-UR-22-30006, 2022.